Processamento de sinais digitais ⇒ Victor C. de L. Guerra, Trabalho DF1 - pt.1

1) $x_2[n] = x_1[(n-4) \bmod 8] \Rightarrow x_2[n] = x_1[n+4]$

$x_2[n] = x_1[n-n_0] \Rightarrow X_2[k] = e^{-j\frac{2\pi}{N}kn_0} X_1[k]; \; n_0 = 4; N = 8$

$X_2[k] = e^{-j\frac{2\pi}{8}k \cdot 4} X_1[k] = e^{-j\pi k} X_1[k]$

2) a) $x[0] = 4; x[1] = 1; x[2] = 2; x[3] = 2$

$$X[k] = \sum_{n=0}^{3} x[n] e^{-j\frac{2\pi}{4}kn} \Rightarrow$$

$\Rightarrow X[0] = 4+1+2+2 = 9; \; X[1] = 4 + e^{-j\frac{2\pi}{4}\cdot 1} + 2e^{-j\frac{2\pi}{4}2} + 2e^{-j\frac{2\pi}{4}\cdot 3} =$

$= 4 + e^{-j\frac{\pi}{2}} + 2e^{-j\pi} + 2e^{-j\frac{3\pi}{2}};$

$X[2] = 4 + e^{-j\frac{\pi 2}{4}\cdot 2} + 2e^{-j\frac{\pi}{4}2\cdot 2} + 2e^{-j\frac{\pi 2}{4}\cdot 3} = 4 + e^{-j\pi} + 2e^{-j\pi 2} + 2e^{-j3\pi}$

$X[3] = 4 + e^{-j\frac{2\pi}{4}\cdot 3} + 2e^{-j\frac{2\pi}{4}\cdot 3\cdot 2} + 2e^{-j\frac{2\pi}{4}\cdot 3\cdot 3} = 4 + e^{-j\frac{3\pi}{2}} + 2e^{-j3\pi} + 2e^{-j\frac{9\pi}{2}}$

b) $y[n] = \sum_{m=0}^{3} x[m]\, x[(n-m) \bmod 4], \; n = 0,1,2,3$

$y[0] = x[0]x[0] + x[1]x[3] + x[2]x[2] + x[3]x[1] = 4\cdot 4 + 1\cdot 2 + 2\cdot 2 + 2\cdot 1 = 16+2+4+2 = 24$

$y[1] = x[0]\cdot x[1] + x[1]x[0] + x[2]x[3] + x[3]x[2] = 4\cdot 1 + 1\cdot 4 + 2\cdot 2 + 2\cdot 2 = 16$

$y[2] = x[0]x[2] + x[1]x[1] + x[2]x[0] + x[3]x[3] = 4\cdot 2 + 1\cdot 1 + 2\cdot 4 + 2\cdot 2 = 21$

$y[3] = x[0]x[3] + x[1]x[2] + x[2]x[1] + x[3]x[0] = 4\cdot 2 + 1\cdot 2 + 2\cdot 1 + 2\cdot 4 = 20$

$y = [24, 16, 21, 20]$

c) $Y[k] = X[k] \cdot X[k] = X[k]^2 \Rightarrow Y[0] = 9^2 = 81;\ Y[1] = (2+j)^2 = 4+4j-1 = 3+4j$

$Y[2] = 3^2 = 9;\ Y[3] = (2-j)^2 = 4-4j-1 = 3-4j \Rightarrow Y = [81, 3+4j, 9, 3-4j]$

$y[n] = \frac{1}{4}\sum_{k=0}^{3} Y[k]\, e^{j\frac{\pi}{2}kn}$; $n = 0,1,2,3$; $e^{j\frac{\pi}{2}} = j$; $e^{j\pi} = -1$; $e^{j\frac{3\pi}{2}} = -j$

$y[0] = \frac{1}{4}\left(81+(3+4j)+9+(3-4j)\right) = \frac{1}{4}(81+3+9+3) = \frac{96}{4} = 24$

$y[1] = \frac{1}{4}\left(81+(3+4j)j + 9(-1) + (3-4j)(-j)\right) = \frac{64}{16} = 16$

$y[2] = \frac{1}{4}\left(81-(3+4j)+9-(3-4j)\right) = \frac{1}{4}(81-3-4j+9-3+4j) = \frac{84}{4} = 21$

$y[3] = \frac{1}{4}\cdot\left(81+(3+4j)(-j) + 9(-1) + (3-4j)j\right) = \frac{80}{4} = 20$

$y[n] = [24, 16, 21, 20]$

d) $L_{lin} = 4+4-1 = 7 \therefore N_{min} = 7$

$n=0 \Rightarrow y_{lin}[0] = 4\cdot 4 = 16$; $n=1 \Rightarrow y_{lin}[1] = 4\cdot 1 + 1\cdot 4 = 8$; $n=2 \Rightarrow y_{lin}[2] = 4\cdot 2 + 1\cdot 1 + 2\cdot 4 = 17$

$n=3 \Rightarrow y_{lin}[3] = 4\cdot 2 + 1\cdot 2 + 2\cdot 1 + 2\cdot 4 = 20$; $n=4 \Rightarrow y_{lin}[4] = 1\cdot 2 + 2\cdot 2 + 2\cdot 1 = 8$; $n=5 \Rightarrow y_{lin}[5] = 2^2 + 2^2 = 8$

$n=6 \Rightarrow y_{lin}[6] = 2\cdot 2 = 4$

3) $x_3[n] = \sum_{k=0}^{7} x_1[k]\cdot x_2[(n-k) \bmod 8] \Rightarrow x_3[2] = \sum_{k=0}^{7} x_1[k]\cdot x_2[(2-k) \bmod 8]$ $k[0,1\ldots$

$x_3[2] = 1\cdot 2 + 1\cdot 3 + 0+0+0+0+0+0+0 = 5 \Rightarrow x[2] = 5$

Processamento de sinais digitais ⇒ Victor E. de L. Guerra, Trabalho DFT - pt. 2

4) $x_1[n] = \sum_{m=-\infty}^{\infty} x[n+4m]$; $n = [0, 1, 2, 3]$

$\Rightarrow x[0] = 1,\ x[1] = 0,\ x[2] = 2,\ x[3] = 2,\ x[4] = b,\ x[5] = 1$

$\Rightarrow x_1[0] = 4,\ x_1[1] = 1,\ x_1[2] = 2,\ x_1[3] = 2$

$\Rightarrow x_1[0] = x[0] + x[4] + x[8] + \cdots = 1 + b + 0 + \cdots = 1 + b \Rightarrow x_1[0] = 4 \Rightarrow 4 = 1 + b \Rightarrow b = 3$

5) a) $x_1[0] = 1,\ x_1[1] = -2,\ x_1[2] = -1,\ x_1[3] = 3$

$x_2[1] = 2,\ x_2[4] = -1,\ x_2[5] = 1$

$y[m] = \sum_{k=0}^{4} x_1[k]\, x_2[(m-k) \bmod 5] \Rightarrow x_2^{(1)} = [1, 2, 0, 0, -1]$

$m=0$: $k=0 \Rightarrow 1 \cdot x_2[0] = 1 \cdot 1 = 1$; $k=1 \Rightarrow -2 \cdot x_2[4] = -2 \cdot (-1) = 2$

$k=2 \Rightarrow -1 \cdot x_2[3] = 0$; $k=3 \Rightarrow -1 \cdot x_2[2] = 0$; $k=4 \Rightarrow 0 \cdot x_2[1] = 0$ $\Rightarrow y[0] = 1+2 = 3$

$m=1$: $k=0 \Rightarrow 1 \cdot x_2[1] = 1 \cdot 2 = 2$; $k=1 \Rightarrow -2 \cdot x_2[0] = -2 \cdot 1 = -2$; $k=2 \Rightarrow -1 \cdot x_2[4] = -1 \cdot (-1) = 1$

$k=3 \Rightarrow 3 \cdot x_2[3] = 0$; $k=4 \Rightarrow 0 \cdot x_2[2] = 0 \Rightarrow y[1] = 2 - 2 + 1 = 1$

$m=2$: $k=0 \Rightarrow 1 \cdot x_2[2] = 0$; $k=1 \Rightarrow -2 \cdot x_2[1] = -4$; $k=2 \Rightarrow -1 \cdot x_2[0] = -1$; $k=3 \Rightarrow 3 \cdot x_2[4] = -3$

$k=4 \Rightarrow 0 \cdot x_2[3] = 0$; $\Rightarrow y[2] = -4 - 1 - 3 = -8$

$m=3$: $k=0 \Rightarrow 1 \cdot x_2[3] = 0$; $k=1 \Rightarrow -2 \cdot x_2[2] = 0$; $k=2 \Rightarrow -1 \cdot x_2[1] = -2$; $k=3 \Rightarrow 3 \cdot x_2[0] = 3 \cdot 1 = 3$

$k=4 \Rightarrow 0 \cdot x_2[4] = 0 \Rightarrow y[3] = -2 + 3 = 1$

$m=4$: $k=0$: $1 \cdot x[4] = -1$; $k=1 \Rightarrow -2 \cdot x[3] = 0$; $k=2 \Rightarrow -1 \cdot x[2] = 0$; $k=3 \Rightarrow 3 \cdot x_2[1] = 6$; $k=4 \Rightarrow 0 \cdot x_2[0] = 0$

$y[4] = -1 + 6 = 5$

$y[m] = [3, 1, -8, 1, 5]$

b) $N \geq L_1 + L_2 - 1 \Rightarrow L_1 = 4,\ L_2 = 6 \Rightarrow L = 4+6+(-1) = 9$

$N_{min} = 9 \Rightarrow y[n] = \sum_{k=0}^{3} x_1[k] \cdot x_2[n-k]$

$n=0$: $y[0] = x_1[0] \cdot x_2[0] = 1 \cdot 0 = 0$

$n=1$: $y[1] = x_1[0]\, x_2[1] + x_1[1]\, x[0] = 1 \cdot 2 + (-2) \cdot 0 = 2$

$n=2$: $y[2] = x_1[0] x_2[2] + x_1[1] x_2[1] + x_1[2] x_2[0] = -2 \cdot 2 = -4$

$n=3$: $y[3] = x_1[0] x_2[3] + x_1[1] x_2[2] + x_1[2] x_2[1] + x_1[3] x_2[0] = -1 \cdot 2 = -2$

$n=4$: $y[4] = x_1[0] x_2[4] + x_1[1] x_2[3] + x_1[2] x_2[2] + x_1[3] x_2[1] + x_1[4] x_2[0] = 1 \cdot (-1) + (3 \cdot 2) = 6 - 1 = 5$

$n=5$: $y[5] = x_1[0] x_2[5] + x_1[1] x_2[4] + x_1[2] x_2[3] + x_1[3] x_2[2] = 1 + 2 = 3$

$n=6$: $y[6] = x_1[1] x_2[5] + x_1[2] x_2[4] + x_1[3] x_2[3] = -2 + 1 = -1$

$n=7$: $y[7] = x_1[2] x_2[5] + x_1[3] x_2[4] = -1 \cdot 3 = -4$

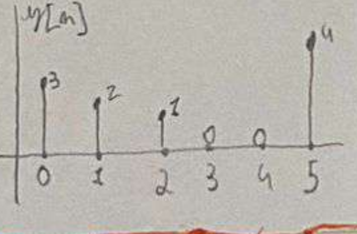
$n=8$: $y[8] = x_1[3] x_2[5] = 3 \cdot 1 = 3$

$y_{lin} = [0, 2, -4, -2, 5, 3, -1, -4, 3]$

6) a) $Y[k] = W_N^{n_0 k} X[k] = e^{-j 2\frac{\pi}{N} \cdot k n_0} X[k] \Rightarrow y[n] = x[(n - n_0) \bmod N]$

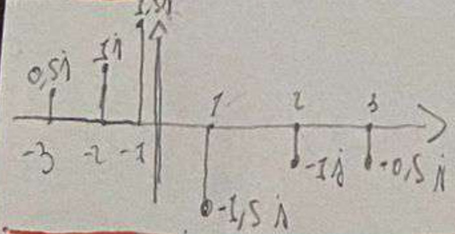
$N = 6,\ n_0 = 5 \Rightarrow y[n] = x[(n-5) \bmod 6] \Rightarrow x[n] = [4, 3, 2, 1, 0, 0]$

$y[0] = x[1] = 3$; $y[1] = x[2] = 2$; $y[2] = x[3] = 1$; $y[3] = x[4] = 0$; $y[4] = x[5] = 0$; $y[5] = x[0] = 4$



b) $w[n] = \frac{x[n] - x^*[-n]}{2j}$ | $x^*[-n] = x[-n]$

$w[n] = x[n] = \frac{x[n] - x[-n]}{2j} = \frac{-j[x[n] - x[-n]]}{2}$



c) $X[k] = 4 + 3e^{-j\frac{\pi 2}{6}k} + 2e^{-j\frac{2\pi}{6}2k} + e^{-j\frac{2\pi}{6}3k}$

$Q[k] = X[2k+1] \Rightarrow Q[k] = 4 + 3(e^{-j\frac{\pi}{3}}) e^{-2\frac{\pi}{3}k} + 2(e^{-2\frac{\pi}{3}j}) e^{-j\frac{2\pi}{3}2k} +$

$+ e^{-j\pi} \Rightarrow q[n] = 3\delta[n] + 3(e^{-j\frac{\pi}{3}}) \delta[n-1]$